

Note 2.6 Matrices

Introduction

In this section, we will learn about the matrices and its use in communications networks and transportation systems.

Definition of Matrix

A **matrix** is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix. The plural of matrix is matrices. A matrix with the same number of rows as columns is called **square**. Two matrices are **equal** if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

Definition of Components of Matrix

Let m and n be positive integers and let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The i th **row** of A is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$, the j th **column** of A is the $m \times 1$ matrix

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

The (i, j) th **element** or **entry** of A is the element a_{ij} , that is, the number in i th row and j th column of A . A convenient shorthand notation for the matrix A is $A = [a_{ij}]$.

Matrix Arithmetic

Definition of Addition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ matrices. the **sum** of A and B , denoted by $A + B$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i, j) th element. That is,

$$A + B = [a_{ij} + b_{ij}]$$

Definition of Multiplication

Let A be an $m \times k$ matrix and B be a $k \times n$ matrix. The **product** of A and B , denoted by AB , is the $m \times n$ matrix with its (i, j) th entry equals to the sum of the products of the corresponding elements from the i th row of A and the j th column of B . That is, if $AB = [c_{ij}]$,

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Note that $AB \neq BA$.

Transposes and Powers of Matrices

Definition of Identity Matrix

The identity matrix of order n is the $n \times n$ matrix $I_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. That is

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

And there exist

$$AI_n = I_m A = A$$

where A is an $m \times n$ matrix.

Power of Square Matrices

When A is an $n \times n$ matrix, we have

$$A^0 = I_n, A^r = \underbrace{AA \cdots A}_{r \text{ times}}$$

Definition of Transpose

Let $A = [a_{ij}]$ be an $m \times n$ matrix, the **transpose** of A , denoted by A^t , is the $n \times m$ matrix obtained by interchanging the rows and columns of A . In other words, if $A^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

Definition of Symmetric

A square matrix A is called symmetric if $A = A^t$. Thus $A = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for all i and j in the domain.

Zero-One Matrices

A matrix all of whose entries are either 0 or 1 is called a **zero-one matrix**, which is often used to represent discrete structures.

Definition of Join and Meet of Zero-One Matrices

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be $m \times n$ zero-one matrices. Then the **join** of A and B is the zero-one matrix with (i, j) th entry $a_{ij} \vee b_{ij}$, denoted by $A \vee B$. The **meet** of A and B is the zero-one matrix with (i, j) th entry $a_{ij} \wedge b_{ij}$. The meet of A and B is denoted by $A \wedge B$.

Definition of Boolean Product

Let $A = [a_{ij}]$ be an $m \times k$ zero-one matrix and $B = [b_{ij}]$ be a $k \times n$ zero-one matrix. Then the Boolean product of A and B , denoted by $A \odot B$, is the $m \times n$ matrix with (i, j) th entry c_{ij} where

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \cdots \vee (a_{ik} \wedge b_{kj})$$

Definition of Boolean Power

Let A be a square zero-one matrix and let r be a positive integer. The r th **Boolean Power** of A is the Boolean product of r factors of A . The r th Boolean product of A is denoted by $A^{[r]}$. Hence

$$A^{[r]} = \underbrace{A \odot A \odot A \odot \cdots \odot A}_{r \text{ times}}$$

We also define $A^{[0]}$ to be I_n .